

Knowledge-Parameter Topology via Reinforcement-Guided Stochastic Spectral Identification in Large Language Models

RL-KPT-SI: A Conceptual Framework for Efficient Discovery of Knowledge-Related Parameter Structures

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August 2026

Abstract

Large Language Models (LLMs) demonstrate remarkable capabilities through large-scale parameterized representations. However, the relationship between semantic knowledge and internal model parameters remains largely unknown.

The Reinforcement-Learned Knowledge-Parameter Topology (RL-KPT) framework proposes that knowledge acquisition in LLMs may involve distributed parameter subspaces rather than isolated parameters, and introduces reinforcement learning mechanisms for selective parameter plasticity during continual learning.

However, an important unresolved question remains:

How can knowledge-related parameter regions be efficiently identified inside extremely large parameter spaces before targeted updating?

This paper proposes Reinforcement-Guided Knowledge-Parameter Topology via Stochastic Spectral Identification (RL-KPT-SI), a conceptual mathematical framework for discovering knowledge-sensitive parameter structures in large language models.

Instead of directly computing complete parameter sensitivities, which is computationally infeasible for modern LLMs, RL-KPT-SI introduces an adaptive exploration mechanism based on reinforcement learning and statistical sampling. The framework treats parameter regions as searchable functional units and learns which regions provide the maximum information about knowledge-related parameter structures.

A knowledge-conditioned Jacobian is introduced to describe the relationship between model responses and parameter perturbations:

$$J_K = \frac{\partial f_\theta(K)}{\partial \theta}$$

Because the full Jacobian is impractical to compute for large models, RL-KPT-SI proposes stochastic approximation through adaptive sampling, random projection, and low-dimensional estimation.

The resulting approximate sensitivity structure is analyzed through spectral methods to identify knowledge-sensitive parameter subspaces:

$$\mathcal{S}_K \subseteq \mathbb{R}^N$$

Based on this representation, this paper introduces the concepts of knowledge-sensitive parameter subspace, spectral knowledge signature, knowledge parameter fingerprint, knowledge similarity geometry, and knowledge interference geometry.

This work does not claim that knowledge localization inside LLM parameters has already been experimentally established. Instead, it provides a mathematically testable research framework for future investigation.

If validated, RL-KPT-SI may provide a foundation for knowledge-aware continual learning, selective parameter updating, and autonomous knowledge expansion mechanisms in future LLM architectures.

1 Introduction

Large Language Models have achieved remarkable performance in natural language understanding, reasoning, coding, and domain-specific tasks. These capabilities emerge from optimization over extremely large parameter spaces.

A conventional formulation represents an LLM as:

$$f_{\theta}(x)$$

where:

$$\theta = \{\theta_1, \theta_2, \dots, \theta_N\}$$

denotes the complete parameter set.

Current training paradigms generally optimize:

$$\theta^* = \arg \min_{\theta} L(\theta)$$

where knowledge acquisition is achieved through global parameter optimization.

Although effective, this paradigm introduces several fundamental limitations:

1. New knowledge acquisition requires expensive retraining or fine-tuning.
2. Parameter updates may interfere with previously acquired knowledge.
3. The internal relationship between semantic knowledge and parameter structures remains unclear.

The central question addressed by this work is:

How is semantic knowledge organized inside the parameter space of large language models?

This paper proposes that knowledge may not correspond to individual parameters, but rather to distributed functional parameter structures.

$$Knowledge \neq Single\ Parameter$$

Instead:

$$Knowledge \rightarrow Distributed\ Parameter\ Subspace$$

This perspective follows the Knowledge-Parameter Topology idea that semantic concepts may correspond to distributed and overlapping regions inside neural parameter space.

Following this work, the RL-KPT-SI framework extends the idea of knowledge-parameter topology by investigating how knowledge-related parameter structures can be identified through reinforcement-guided stochastic spectral analysis.

2 Relation to RL-KPT: Previous Work and Research Extension

The RL-KPT framework [?] proposed that continual learning in LLMs requires selective parameter plasticity rather than unrestricted global parameter modification.

The previous work introduced the concept of Knowledge-Parameter Topology, suggesting that knowledge acquisition may correspond to structured parameter regions instead of uniformly distributed parameter changes.

Instead of updating all parameters:

$$\theta_{t+1} = \theta_t + \Delta\theta$$

RL-KPT introduces a parameter selection mechanism:

$$\theta_{t+1} = \theta_t + M_t \odot \Delta\theta_t$$

where:

$$M_t$$

represents the parameter update mask.

The mask is generated through a reinforcement learning policy:

$$M_t = \pi_\phi(S_t, K_t)$$

where:

- S_t represents the model state;
- K_t represents the incoming knowledge;
- π_ϕ represents the learned routing policy.

However, RL-KPT introduces an important open problem:

Before updating knowledge – related parameters, how can these parameters be identified?

RL-KPT-SI addresses this identification problem.

Therefore:

RL – KPT : Knowledge Update Routing

while:

RL – KPT – SI : Knowledge Parameter Identification

Together they form:

$$\textit{Knowledge} \rightarrow \textit{Identification} \rightarrow \textit{Localization} \rightarrow \textit{Selective Updating}$$

RL-KPT-SI extends the previous RL-KPT framework from knowledge updating to knowledge identification.

While RL-KPT focuses on how reinforcement learning can regulate parameter plasticity during continual knowledge acquisition, RL-KPT-SI investigates the prior question of where knowledge-related parameter structures exist inside large language models.

Therefore, the relationship between the two frameworks can be summarized as:

RL – KPT : Knowledge Update Mechanism

RL – KPT – SI : Knowledge Parameter Identification

Together, they form a complete research direction:

$$\textit{Knowledge} \rightarrow \textit{Identification} \rightarrow \textit{Localization} \rightarrow \textit{Selective Updating}$$

The RL-KPT framework serves as the theoretical foundation for the selective plasticity mechanism, while RL-KPT-SI provides a possible mathematical approach for discovering knowledge-sensitive parameter structures before updating.

2.1 Research Evolution from RL-KPT to RL-KPT-SI

The research evolution can be summarized as:

RL – KPT : Where and how should parameters be updated?

RL – KPT – SI : How can knowledge – related parameter structures be identified?

The two frameworks address different stages of knowledge evolution in large language models:

Knowledge → Parameter Identification → Parameter Localization → Selective Adaptation

Therefore, RL-KPT-SI should be considered as an extension of the RL-KPT framework rather than an independent hypothesis.

3 Research Hypothesis

The central hypothesis of RL-KPT-SI is:

Knowledge states induce structured parameter – response patterns

When a model processes a knowledge-bearing input, perturbations in different parameter regions may produce different effects on the model response.

Therefore, the relationship between knowledge and parameters can be studied through:

$$K \rightarrow f_{\theta}(K) \rightarrow \frac{\partial f_{\theta}(K)}{\partial \theta}$$

The hypothesis does not assume that each fact is stored in an isolated parameter. Instead, knowledge may occupy a distributed subspace:

$$\mathcal{S}_K \subseteq \mathbb{R}^N$$

where:

$$N$$

is the total number of model parameters.

Different knowledge concepts may have overlapping structures:

$$\mathcal{S}_{K_1} \cap \mathcal{S}_{K_2} \neq \emptyset$$

Such overlap may explain knowledge composition and interference phenomena in continual learning.

4 Knowledge-Conditioned Parameter Response

Let:

$$K_i$$

represent a knowledge item.

The model response is:

$$y_i = f_\theta(K_i)$$

A local parameter perturbation:

$$\delta\theta$$

causes:

$$\delta y_i \approx J_i \delta\theta$$

where:

$$J_i = \frac{\partial y_i}{\partial \theta}$$

is the knowledge-conditioned Jacobian.

The Jacobian represents:

$$\boxed{Parameter\ Perturbation \rightarrow Knowledge\ Response}$$

Therefore, the Jacobian provides a mathematical bridge between semantic knowledge and parameter geometry.

5 Computational Challenge

For modern LLMs:

$$N \geq 10^{10}$$

or even:

$$N \geq 10^{12}$$

The complete Jacobian:

$$J_i$$

becomes computationally impossible to store and analyze.

Therefore, RL-KPT-SI does not attempt full parameter enumeration.

Instead, it proposes:

$$\boxed{Full\ Parameter\ Analysis \rightarrow Adaptive\ Statistical\ Identification}$$

The goal is not to measure every parameter.

The goal is:

$$\boxed{Identify\ the\ most\ informative\ knowledge - related\ parameter\ regions}$$

6 Parameter Topology Representation

Instead of individual parameters:

$$\theta_1, \theta_2, \dots, \theta_N$$

RL-KPT-SI defines functional parameter regions:

$$P = \{P_1, P_2, \dots, P_q\}$$

where each region may represent:

- Transformer layer block;
- Attention head;
- MLP channel;
- Neuron cluster;
- Low-rank parameter subspace;
- Cross-layer functional structure.

Since:

$$q \ll N$$

the identification problem becomes a search problem over functional regions rather than individual parameters.

$$Find: P_k$$

such that:

$$P_k$$

contains maximum information about knowledge representation.

7 Reinforcement Learning Guided Parameter Sampling

The key idea of RL-KPT-SI is that reinforcement learning is not only used for parameter updating, but can also be used as an adaptive exploration mechanism for discovering knowledge-related parameter structures.

Traditional parameter analysis assumes that all parameters should be examined:

$$\theta = \{\theta_1, \theta_2, \dots, \theta_N\}$$

However, RL-KPT-SI assumes that knowledge-related information is sparsely distributed inside a much larger parameter space.

Therefore, the identification problem can be formulated as:

$$\max_{\pi} I(K; P)$$

where:

- K represents the knowledge state;

- P represents the explored parameter regions;
- $I(K; P)$ represents the information obtained about knowledge structure from parameter observations.

The objective is to discover parameter regions that maximize knowledge information while minimizing computational cost.

7.1 RL Exploration State

The reinforcement learning agent observes:

$$S_t = (K_t, H_t, A_t)$$

where:

- K_t represents the current knowledge representation;
- H_t represents hidden activation states produced by the model;
- A_t represents previously explored parameter regions.

The action space is defined as selecting a parameter region:

$$a_t = P_i$$

where:

$$P_i \in \{P_1, P_2, \dots, P_q\}$$

The policy network learns:

$$a_t = \pi_\phi(S_t)$$

which determines the next parameter region to investigate.

7.2 Information-Based Reward

Unlike conventional reinforcement learning objectives, the reward is not based on immediate task performance.

Instead, RL-KPT-SI defines an information-oriented reward:

$$R_t = \Delta I(P_i, K) - \lambda C(P_i)$$

where:

- $\Delta I(P_i, K)$ represents the knowledge information gain;
- $C(P_i)$ represents computational cost;
- λ controls the exploration efficiency trade-off.

The optimization objective becomes:

$$\max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t R_t \right]$$

Through repeated exploration, the RL agent learns which parameter regions provide the most useful information for constructing the knowledge-parameter topology.

8 Hierarchical Parameter Sampling

Modern Transformer models contain different structural levels:

$$\Theta = \{\Theta_L, \Theta_A, \Theta_M\}$$

where:

- Θ_L represents layer-level structures;
- Θ_A represents attention structures;
- Θ_M represents MLP structures.

RL-KPT-SI proposes hierarchical exploration:

$$Layer \rightarrow Component \rightarrow Subspace \rightarrow Parameter$$

Instead of directly searching billions of parameters, the agent first identifies high-level functional regions.

For example:

$$P_i = Layer_8 + AttentionHead_{12} + MLPChannel_{320}$$

may represent a candidate knowledge-sensitive region.

This hierarchical mechanism reduces search complexity from:

$$O(N)$$

to:

$$O(q)$$

where:

$$q \ll N$$

represents the number of functional regions.

9 Approximate Knowledge Jacobian

Although the ideal knowledge-conditioned Jacobian is defined as:

$$J_K = \frac{\partial f_\theta(K)}{\partial \theta}$$

direct calculation is impossible for large-scale models.

RL-KPT-SI therefore introduces an approximate formulation:

$$\hat{J}_K \approx J_K M$$

where:

$$M$$

is an adaptive sampling operator generated by the RL exploration policy.

The sampling operator:

$$M = [m_1, m_2, \dots, m_q]$$

selects informative parameter regions.

Therefore:

$$\hat{J}_K = \frac{\partial f_\theta(K)}{\partial \theta} M$$

represents the knowledge response of selected parameter regions.

The objective becomes:

$$\min_M \|J_K - \hat{J}_K\|$$

under a computational budget constraint.

The key hypothesis is:

A small number of informative parameter regions may preserve most knowledge – related response information

10 Random Projection and Low-Dimensional Approximation

Even after adaptive sampling, the Jacobian may remain large.

Therefore RL-KPT-SI introduces stochastic dimensionality reduction.

A random projection matrix:

$$R \in \mathbb{R}^{N \times r}$$

is applied:

$$Z_K = J_K R$$

where:

$$r \ll N$$

According to high-dimensional random projection principles, the projected space may preserve important geometric relationships.

The approximate sensitivity matrix becomes:

$$\hat{G}_K = Z_K^T Z_K$$

instead of:

$$G_K = J_K^T J_K$$

This transforms an impossible:

$$N \times N$$

matrix problem into:

$$r \times r$$

spectral analysis.

10.1 Low-Rank Knowledge Representation

The approximate Jacobian may also be decomposed:

$$\hat{J}_K \approx U_r \Sigma_r V_r^T$$

where:

- U_r represents response directions;
- Σ_r represents sensitivity strength;
- V_r represents important parameter directions.

The dominant parameter directions:

$$V_r$$

define the estimated knowledge-sensitive parameter subspace:

$$\hat{\mathcal{S}}_K = \text{span}(V_r)$$

This provides a computationally feasible approximation of the ideal knowledge topology.

11 Spectral Knowledge Identification

After obtaining the approximate sensitivity matrix:

$$\hat{G}_K = \hat{J}_K^T \hat{J}_K$$

spectral decomposition is applied:

$$\hat{G}_K = V \Lambda V^T$$

where:

$$\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_r)$$

represents sensitivity energy distribution.

The dominant eigenvectors:

$$V_1, V_2, \dots, V_k$$

define the knowledge-sensitive parameter subspace:

$$\boxed{\hat{\mathcal{S}}_K = \text{span}(V_1, V_2, \dots, V_k)}$$

where k is selected according to an effective spectral criterion.

12 Effective Knowledge Rank

The effective rank of knowledge is defined as:

$$r_{eff}(K)$$

A possible criterion is:

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^r \lambda_i} \geq \tau$$

where:

$$0 < \tau < 1$$

represents the retained spectral information ratio.

The effective rank provides a possible measurement of knowledge distribution:

- Low effective rank:
knowledge may be concentrated in a small functional subspace.
- High effective rank:
knowledge may be distributed across broader parameter structures.

13 Spectral Knowledge Signature

For each knowledge item K_i , define its spectral signature:

$$\Sigma(K_i) = \{\lambda_1, \lambda_2, \dots, \lambda_k\}$$

A normalized form:

$$p_i = \frac{\lambda_i}{\sum_j \lambda_j}$$

creates a spectral distribution:

$$P_K = [p_1, p_2, \dots, p_k]$$

The central hypothesis is:

Different knowledge structures may produce different spectral signatures.

This signature does not represent semantic meaning directly.

Instead, it represents the geometric response pattern induced by knowledge in parameter space.

14 Knowledge Parameter Fingerprint

The complete knowledge parameter fingerprint is defined as:

$$\Phi(K) = [\Sigma(K), r_{eff}, \mathcal{S}_K, L_K, C_K]$$

where:

- $\Sigma(K)$ represents the spectral signature;

- r_{eff} represents effective knowledge rank;
- $\mathcal{S}_K$ represents the sensitive parameter subspace;
- L_K represents layer/component distribution;
- C_K represents cross-component coupling.

The fingerprint provides a possible representation of knowledge organization inside the model parameter space.

15 Knowledge Similarity Geometry

Given two knowledge items:

$$K_i, K_j$$

their parameter-space similarity can be defined through their subspace relationship:

$$D(K_i, K_j) = d(\mathcal{S}_{K_i}, \mathcal{S}_{K_j})$$

Possible geometric measures include:

- Principal angle;
- Projection overlap;
- Canonical correlation;
- Subspace distance.

A possible overlap measurement:

$$O_{ij} = \frac{Tr(P_i P_j)}{\min(r_i, r_j)}$$

where:

$$P_i, P_j$$

are projection matrices of corresponding knowledge subspaces.

This provides a possible mathematical description of knowledge relatedness in parameter space.

16 Knowledge Interference Geometry

A major challenge in continual learning is knowledge interference.

When new knowledge is introduced, parameter updates may modify structures that support previously acquired knowledge.

RL-KPT-SI proposes that interference may be related to geometric relationships between knowledge-sensitive parameter subspaces.

For two knowledge concepts:

$$K_i, K_j$$

their interference relationship may be analyzed through:

$$\mathcal{I}_{ij} = g(\mathcal{S}_{K_i}, \mathcal{S}_{K_j})$$

where:

$$\mathcal{S}_{K_i}$$

and:

$$\mathcal{S}_{K_j}$$

represent their corresponding parameter-sensitive subspaces.

16.1 Update Direction Interaction

Suppose two knowledge updates generate parameter changes:

$$\Delta\theta_i$$

and:

$$\Delta\theta_j$$

Their directional relationship can be measured by:

$$C_{ij} = \frac{\Delta\theta_i^T \Delta\theta_j}{\|\Delta\theta_i\| \|\Delta\theta_j\|}$$

Possible interpretations:

$$C_{ij} > 0$$

indicates compatible update directions;

$$C_{ij} \approx 0$$

indicates weak interaction;

$$C_{ij} < 0$$

indicates potentially conflicting update directions.

However, parameter update conflict is not necessarily identical to semantic conflict. Therefore, RL-KPT-SI defines a broader concept:

$$\boxed{\textit{Knowledge Interference Geometry}}$$

which considers both:

1. parameter subspace overlap;
2. update direction interaction;
3. spectral structure similarity.

17 Knowledge Composition and Topological Structure

Knowledge in real-world applications is usually compositional.

For example, an engineering knowledge concept may involve:

$$K_R = K_{physics} + K_{mechanics} + K_{control}$$

The corresponding parameter topology may not require an independent parameter region. Instead:

$$\mathcal{S}_{K_R}$$

may be constructed from existing knowledge-sensitive subspaces. A simple formulation:

$$\mathcal{S}_{K_R} \approx \mathcal{S}_{physics} + \mathcal{S}_{mechanics} + \mathcal{S}_{control}$$

However, because Transformer representations are nonlinear, a more general formulation is:

$$\mathcal{S}_{K_R} = F_S(\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_n)$$

where:

$$F_S$$

represents nonlinear knowledge composition.

This leads to an important research question:

Can complex knowledge be predicted through combinations of existing parameter topologies?

18 Unified RL-KPT-SI Architecture

RL-KPT and RL-KPT-SI represent two complementary components.

RL-KPT focuses on:

$$Knowledge \rightarrow Selective Updating$$

while RL-KPT-SI focuses on:

$$Knowledge \rightarrow Parameter Identification$$

Together they form:

$$Knowledge \rightarrow Identification \rightarrow Localization \rightarrow Reinforcement Learning \rightarrow Selective Plasticity$$

The complete architecture can be described as:

$$K \rightarrow \Phi(K) \rightarrow \mathcal{S}_K \rightarrow M \rightarrow \Delta\theta$$

where:

- K represents incoming knowledge;
- $\Phi(K)$ represents the knowledge parameter fingerprint;

- $\mathcal{S}_K$ represents the identified parameter subspace;
- M represents the update mask;
- $\Delta\theta$ represents parameter modification.

18.1 Knowledge Identification Layer

The identification layer estimates:

$$\Phi(K)$$

through:

$$K \rightarrow RL \text{ Sampling} \rightarrow \hat{J}_K \rightarrow \hat{G}_K \rightarrow \Phi(K)$$

where:

$$\hat{G}_K = \hat{J}_K^T \hat{J}_K$$

represents the approximate knowledge sensitivity matrix.

18.2 Knowledge Update Layer

After identification:

$$\Phi(K) \rightarrow M_K$$

the RL-KPT mechanism performs:

$$\theta_{t+1} = \theta_t + M_K \odot \Delta\theta_t$$

Therefore, the model does not require unrestricted global parameter plasticity. Instead:

$$Knowledge \text{ identification enables selective plasticity}$$

19 Research Hypotheses

RL-KPT-SI proposes several experimentally testable hypotheses.

19.1 H1: Knowledge-Conditioned Parameter Structure

Different knowledge states produce structured parameter response patterns rather than random sensitivity distributions.

19.2 H2: Effective Knowledge Subspace

The dominant response of an LLM to a knowledge concept can be represented by a lower-dimensional parameter-sensitive subspace.

19.3 H3: Spectral Knowledge Signature

Different knowledge domains may produce distinguishable spectral signatures in parameter-response space.

19.4 H4: Parameter-Space Knowledge Similarity

Semantically related knowledge may exhibit greater similarity between their parameter-sensitive subspaces.

19.5 H5: Knowledge Interference Geometry

The degree of parameter-space overlap may correlate with continual learning interference.

19.6 H6: RL-Based Efficient Identification

Adaptive reinforcement learning sampling can reduce the computational cost of identifying knowledge-related parameter structures.

20 Proposed Experimental Research Program

The following experiments are proposed to investigate the hypotheses.

20.1 Experiment 1: Jacobian Structure Analysis

A small-scale Transformer model can be used to estimate:

$$J_K$$

for controlled knowledge categories.

The objective is to investigate whether knowledge-conditioned sensitivity patterns exhibit stable structures.

20.2 Experiment 2: Approximate Identification

Compare:

$$J_K$$

with:

$$\hat{J}_K$$

obtained through RL-guided sampling.

Evaluation criteria:

- information retention;
- computational reduction;
- topology reconstruction accuracy.

20.3 Experiment 3: Spectral Knowledge Classification

Construct:

$$\Phi(K)$$

for different knowledge categories:

- mathematics;

- physics;
- engineering;
- programming;
- biology;
- humanities.

Test whether knowledge categories can be distinguished using parameter fingerprints rather than raw text.

20.4 Experiment 4: Knowledge Similarity Analysis

Measure:

$$D(\mathcal{S}_{K_i}, \mathcal{S}_{K_j})$$

and compare with semantic similarity.

The goal is to determine whether semantic relationships correspond to parameter-space geometry.

20.5 Experiment 5: Continual Learning Interference

Sequentially introduce knowledge:

$$K_1 \rightarrow K_2 \rightarrow K_3$$

Measure:

$$\mathcal{I}_{ij}$$

and compare with observed forgetting.

20.6 Experiment 6: Reverse Knowledge Identification

The most important experiment is reverse inference.

Given:

$$\Phi(K_x)$$

without observing the original knowledge text:

$$\Phi(K_x) \rightarrow \textit{Knowledge Category}$$

The experiment tests whether parameter-response geometry contains information about underlying knowledge.

21 Limitations

RL-KPT-SI is a conceptual framework and does not claim that all hypotheses have already been experimentally verified.

Several limitations remain:

1. Knowledge may not always correspond to uniquely identifiable parameter structures.
2. Sensitivity does not necessarily imply causal storage.
3. Jacobian-based analysis may capture influence rather than actual knowledge location.
4. Spectral signatures may depend on model architecture, scale, and training procedure.
5. The relationship between semantic knowledge similarity and parameter geometry requires empirical validation.

These limitations define future research directions rather than invalidate the framework.

22 Discussion

The central idea of RL-KPT-SI is that knowledge inside large language models may be studied through parameter-response geometry rather than direct parameter inspection.

The proposed transformation is:

$$\textit{Parameter Value} \neq \textit{Knowledge}$$

Instead:

$$\textit{Knowledge} \rightarrow \textit{Parameter Response} \rightarrow \textit{Geometric Structure}$$

This perspective provides a possible mathematical language for studying:

- continual learning;
- knowledge editing;
- parameter-efficient adaptation;
- autonomous model expansion;
- knowledge-aware routing.

If future experiments validate the proposed hypotheses, LLMs may be able to analyze their own knowledge organization and perform more selective and efficient learning.

23 Conclusion

This paper proposes RL-KPT-SI, a conceptual framework for discovering knowledge-related parameter topology in large language models.

The framework combines:

$\textit{Reinforcement Learning} + \textit{Statistical Sampling} + \textit{Jacobian Analysis} + \textit{Spectral Identification}$

Instead of directly analyzing all parameters, RL-KPT-SI proposes an adaptive exploration mechanism that identifies informative parameter regions and constructs approximate knowledge-sensitive subspaces.

The proposed framework introduces:

Knowledge Parameter Fingerprint

as a possible representation of how semantic knowledge is organized in parameter space.

The ultimate research question is:

Can the hidden knowledge structure of an LLM be identified through its parameter – response geometry?

RL-KPT-SI does not claim to have solved this problem.

Rather, it provides a mathematical hypothesis and research direction for investigating the internal organization of future adaptive intelligent systems.

23.1 Future Research Direction

RL-KPT-SI represents the second stage of a broader research direction.

The first stage, RL-KPT, investigates adaptive knowledge acquisition through reinforcement-guided parameter plasticity.

The current work investigates the inverse problem:

Knowledge $\rightarrow$ Parameter Structure

Future research may combine both directions into a unified adaptive language model architecture:

Knowledge Discovery + Knowledge Identification + Knowledge Updating

Such an architecture may provide a possible foundation for future continual learning systems capable of autonomous knowledge expansion.

Acknowledgement

This work is developed from the author’s previous research direction on Reinforcement Learning Guided Knowledge-Parameter Topology (RL-KPT).

The author thanks researchers in the fields of large language models, continual learning, neural network interpretability, and reinforcement learning whose work provides important foundations for exploring the relationship between knowledge and parameter structures.

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